

Spenny Piggy – FX & Currency Implementation Specification

1. Core Decision

Spenny Piggy will NOT monetise FX. All supporters are charged in the creator's listing currency. No platform-level currency conversion is performed.

2. Currency Rules

- `charge_currency` MUST equal `creator.currency`.
- `display_currency` is UI-only and must never affect Stripe charge currency.
- No FX markup is applied.
- No conversion is performed at platform level.
- Stripe and card issuers handle any cross-border conversion.

3. Gross-Up Formula (Unchanged)

$\text{PriceWithVAT} = \text{ListedPrice} + (\text{ListedPrice} * \text{VATRate})$

$\text{TotalSupporterPays} = \text{ceil}(\text{PriceWithVAT} + \text{FixedFees}) / (1 - \text{TotalPercentageFees})$

All calculations must be performed in minor currency units (cents/pence). Always round UP using `ceil()` to prevent underpayment to creators.

4. Stripe PaymentIntent Configuration

```
createPaymentIntent({
  amount: grossed_up_total_minor_units,
  currency: creator.currency,
  metadata: {
    creator_id: creator.id,
    creator_currency: creator.currency,
    display_currency: selected_display_currency,
    vat_rate: creator.vat_rate
  }
})
```

5. Display Currency (Visual Only)

Display currency is for user interface estimation only. It must not affect charge currency or gross-up calculations.

```
display_amount = listed_price * cached_fx_rate
```

UI Example:

\$100.00 USD

≈ £78.42 GBP (estimated)

Final charge will be in USD.

6. Required Database Fields

- creator_currency
- charge_currency
- display_currency
- stripe_fee_actual
- stripe_fee_expected
- supporter_country
- card_country
- fee_variance

7. Refund Handling

Refunds must occur in the same currency as the original charge. No FX recalculation is required. Fee reversals should be proportional and logged for reconciliation.